

Restricting and Sorting Data

1. Display the Ename, Sal, Comm from EMP table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause.

```
SQL> SELECT ENAME, SAL, COMM
2 FROM EMP
3 WHERE COMM IS NOT NULL
4 ORDER BY 2 DESC, 3 DESC;
```

ENAME	SAL	COMM
ALLEN	1600	300
TURNER	1500	0
MARTIN	1250	1400
WARD	1250	500

2. The HR department needs a query to display all unique job codes from the EMP table.

```
SQL> SELECT DISTINCT JOB
2 FROM EMP;
```

JOB
PRESIDENT
MANAGER
ANALYST
CLERK
SALESMAN

3. The HR department has requested a report of all employees and their job IDs. Display the **last name concatenated with the job ID** (separated by a comma and space) and name the column **Employee and Title** by giving **Column Alias**.

```
SQL> select empno as "Emp#", ename as "Employee", job as "job", hiredate as "hire date" from emp;
```

Emp#	Employee	job	hire date
7839	KING	PRESIDENT	17-NOV-81
7698	BLAKE	MANAGER	01-MAY-81
7782	CLARK	MANAGER	09-JUN-81
7566	JONES	MANAGER	02-APR-81
7788	SCOTT	ANALYST	13-JUL-87
7902	FORD	ANALYST	03-DEC-81
7369	SMITH	CLERK	17-DEC-80
7499	ALLEN	SALESMAN	20-FEB-81
7521	WARD	SALESMAN	22-FEB-81
7654	MARTIN	SALESMAN	28-SEP-81

4. The HR department has requested a report of all employees and their job IDs. Display the **last name concatenated with the job ID** (separated by a comma and space) and name the column **Employee and Title** by giving **Column Alias**.

```
SQL> SELECT ENAME || ',' || JOB || ',' || HIREDATE || ',' || MGR AS THE_OUTPUT
2 FROM emp;

THE_OUTPUT
-----
KING,PRESIDENT,17-NOV-81,
BLAKE,MANAGER,01-MAY-81,7839
CLARK,MANAGER,09-JUN-81,7839
JONES,MANAGER,02-APR-81,7839
SCOTT,ANALYST,13-JUL-87,7566
FORD,ANALYST,03-DEC-81,7566
SMITH,CLERK,17-DEC-80,7902
ALLEN,SALESMAN,20-FEB-81,7698
WARD,SALESMAN,22-FEB-81,7698
MARTIN,SALESMAN,28-SEP-81,7698
TURNER,SALESMAN,08-SEP-81,7698
ADAMS,CLERK,23-MAY-87,7788
JAMES,CLERK,03-DEC-81,7698
MILLER,CLERK,23-JAN-82,7782

14 rows selected.
```

5. To familiarize yourself with the data in the **EMP** table, create a query to display all the data from that table. Separate each column output by a comma. **ENAME, JOB, HIREDATE, MGR**. Name the column title **THE_OUTPUT**.

```
SQL> SELECT ENAME, JOB, HIREDATE, MGR
2 FROM EMP;

ENAME                                JOB                                HIREDATE                                MGR
-----                                -
KING                                PRESIDENT                                17-NOV-81
BLAKE                                MANAGER                                01-MAY-81                                7839
CLARK                                MANAGER                                09-JUN-81                                7839
JONES                                MANAGER                                02-APR-81                                7839
SCOTT                                ANALYST                                13-JUL-87                                7566
FORD                                ANALYST                                03-DEC-81                                7566
SMITH                                CLERK                                17-DEC-80                                7902
ALLEN                                SALESMAN                                20-FEB-81                                7698
WARD                                SALESMAN                                22-FEB-81                                7698
MARTIN                                SALESMAN                                28-SEP-81                                7698
TURNER                                SALESMAN                                08-SEP-81                                7698
ADAMS                                CLERK                                23-MAY-87                                7788
JAMES                                CLERK                                03-DEC-81                                7698
MILLER                                CLERK                                23-JAN-82                                7782

14 rows selected.
```

6. Create a report to display the **ename, job, and hiredate** for the employees whose name is **SCOTT** or **TURNER**. Order the query in **ascending order by hiredate**.

```
SQL> SELECT ENAME, JOB, HIREDATE FROM EMP WHERE ENAME IN ('SCOTT','TURNER') ORDER BY HIREDATE ASC;

ENAME                                JOB                                HIREDATE
-----                                -
TURNER                                SALESMAN                                08-SEP-81
SCOTT                                ANALYST                                13-JUL-87
```

7. Display the **ename** and **department number** of all employees in departments **20** or **30** in **ascending alphabetical order by ename**.

```
SQL> SELECT ENAME, DEPTNO FROM EMP WHERE DEPTNO IN (20,30) ORDER BY ENAME ASC;

ENAME                                DEPTNO
-----                                -
ADAMS                                20
ALLEN                                30
BLAKE                                30
FORD                                20
JAMES                                30
JONES                                20
MARTIN                                30
SCOTT                                20
SMITH                                20
TURNER                                30
WARD                                30

11 rows selected.
```

8. Modify the previous query to display the **last name** and **salary** of employees who earn **between 2000 and 3000** and are in department **20** or **30**. Label the columns **Employee** and **Monthly Salary**, respectively, using **Column Alias**.

```
SQL> SELECT ENAME AS "Employee", SAL AS "Monthly Salary" FROM EMP WHERE SAL BETWEEN 2000 AND 3000 AND DEPTNO IN (20,30);
```

Employee	Monthly Salary
BLAKE	2850
JONES	2975
SCOTT	3000
FORD	3000

9. The HR department needs a report that displays the **last name** and **hire date** for all employees who were **hired in 1981**.

```
SQL> SELECT ENAME, HIREDATE FROM EMP WHERE HIREDATE BETWEEN TO_DATE('01-JAN-1981','DD-MON-YYYY') AND TO_DATE('31-DEC-1981','DD-MON-YY
YY');
```

ENAME	HIREDATE
KING	17-NOV-81
BLAKE	01-MAY-81
CLARK	09-JUN-81
JONES	02-APR-81
FORD	03-DEC-81
ALLEN	20-FEB-81
WARD	22-FEB-81
MARTIN	28-SEP-81
TURNER	08-SEP-81
JAMES	03-DEC-81

10 rows selected.

10. Display the **Ename** and **Sal** of employees who earn **more than an amount the user specifies after a Prompt**.

```
SQL> SELECT ENAME, SAL FROM EMP WHERE SAL > &SALARY;
Enter value for salary: 2000
old 1: SELECT ENAME, SAL FROM EMP WHERE SAL > &SALARY
new 1: SELECT ENAME, SAL FROM EMP WHERE SAL > 2000
```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000

6 rows selected.

11. Create a report to display the **last name** and **job title** of all employees who **do not have a manager**.

```
SQL> SELECT ENAME, JOB FROM EMP WHERE MGR IS NULL;
```

ENAME	JOB
KING	PRESIDENT

12. Create a query that prompts the user for **Manager ID** and generates **EMPNO**, **ENAME**, **SAL**, **DEPTNO**. The user should have the ability to **sort the records on a selected column**.

```
SQL> SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN;
Enter value for mgr: 7698
Enter value for column: ename
old 1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN
new 1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=7698 ORDER BY ename
```

EMPNO	ENAME	SAL	DEPTNO
7499	ALLEN	1600	30
7900	JAMES	950	30
7654	MARTIN	1250	30
7844	TURNER	1500	30
7521	WARD	1250	30

13. The HR department wants to run reports based on a manager. Create a query that prompts the user for a **MGR** and generates the **empno**, **ename**, **salary**, and **department** for that manager's employees. The HR department wants the ability to **sort the report on a selected column**.

```
SQL> SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN;
Enter value for mgr: 7698
Enter value for column: ENAME
old 1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=&MGR ORDER BY &COLUMN
new 1: SELECT EMPNO, ENAME, SAL, DEPTNO FROM EMP WHERE MGR=7698 ORDER BY ENAME
```

EMPNO	ENAME	SAL	DEPTNO
7499	ALLEN	1600	30
7900	JAMES	950	30
7654	MARTIN	1250	30
7844	TURNER	1500	30
7521	WARD	1250	30

14. Display all employee **last names** in which the **third letter of the name is A**.

```
SQL> SELECT ENAME FROM EMP WHERE ENAME LIKE '__A%';
```

ENAME
BLAKE
CLARK
ADAMS

15. Display the **last name** of all employees who have **both an A and an S** in their **ename**.

```
SQL> SELECT ENAME FROM EMP WHERE ENAME LIKE '%A%' AND ENAME LIKE '%S%';
```

ENAME
ADAMS
JAMES

16. Display the **Ename, Job, Sal** for all employees whose **jobs are CLERK** and whose **salary is 800 or 950 or 1300**.

```
SQL> SELECT ENAME, JOB, SAL FROM EMP WHERE JOB='CLERK' AND SAL IN (800,950,1300);
```

ENAME	JOB	SAL
SMITH	CLERK	800
JAMES	CLERK	950
MILLER	CLERK	1300